

Ephedra sinica (Ephedraceae)

Ephedra, Ma Huang (Chinese)

Ephedra is a strongly stimulant, acrid-tasting herb that has a central place in Chinese and other herbal traditions. According to legend, the bodyguards of Genghis Khan, threatened with beheading if they fell asleep on sentry duty, used to take a tea containing ephedra to stay alert. Today, ephedra is used in the West and in China for problems ranging from chills and fevers to asthma and hay fever.



Ma Huang was found in a Middle Eastern neolithic grave, indicating that it may have been used as a medicine 60,000 years ago.

Habitat & Cultivation

Native to northern China and Inner Mongolia, ephedra often grows in desert areas. It is propagated from seed in autumn or by root division in autumn or spring and needs well-drained soil. The stems are gathered throughout the year and dried.

Related Species

Other *Ephedra* species with similar medicinal properties to ephedra grow throughout the northern hemisphere. In North America, related species were used to treat fevers and relieve kidney pain, while in India, *Ephedra* species were taken for asthma, hay fever, and rheumatism.

Key Constituents

- Protoalkaloids (ephedrine, pseudoephedrine)
- Tannins
- Saponin
- Flavone
- Volatile oil

Key Actions

- Increases sweating
- Dilates the bronchioles (small airways in the lungs)
- Dries mucous membranes
- Diuretic
- Stimulant
- Raises blood pressure

Research

■ **Active constituents** Most of the active constituents mimic the effect of adrenaline within the body, increasing alertness. Ephedrine, extracted originally from ephedra, was first synthesized in 1927 and is used as a decongestant and antiasthmatic.

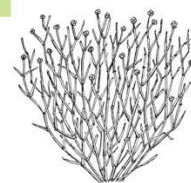
■ **Whole herb** When used at the correct dosage, the whole herb has significant therapeutic effects—including dilating the bronchial airways—and a very low incidence of side effects.

■ **Ephedra abuse** ephedra has been banned in the U.S. due to frequent abuse of the herb, as an amphetamine-type “high” and, at excessive dosage, in weight-loss regimens. At therapeutic dosage, and with professional guidance, ephedra rarely gives rise to side effects.

Traditional & Current Uses

■ **Historical uses** Traditionally, Zen monks used ephedra to promote calm concentration during meditation.

■ **Chinese herb** In China, ephedra is popular for chills and fevers, coughs and wheezing, and in combination with rehmannia (*Rehmannia glutinosa*, p. 125) it is given to treat kidney yin deficiency (see pp. 40–41).

**Ephedra**

An evergreen shrub growing to 20 in (50 cm), with long, narrow sprawling stems and tiny leaves.

Parts Used

Stems are collected throughout the year. *Ma Huang* is best known in medicine for the ephedrine that it contains.



Dried stems



Fresh stems

Key Preparations & Their Uses

■ **Cautions** Take only under professional supervision. Do not take if suffering from angina, glaucoma, high blood pressure, enlarged prostate gland, or overactive thyroid gland. Ephedra occasionally causes side effects, including headaches, tremors, and insomnia. Restricted herb in some countries.



Decoction is prescribed by herbalists for asthma.



Powder is used by the Chinese to treat kidney energy deficiency.



Tincture is used in treatments to alleviate the aches and pains of rheumatism.

Current Western uses

Ephedra is used principally in current Western herbal medicine as a treatment for asthma and hay fever,

and for the acute onset of colds and flu. It also helps to raise blood pressure, cool fevers, and to alleviate rheumatism.